

Stochastic Modeling of Tumor Escape Under Immune Aging and Immune Decline

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Introduction: Cancer progression depends not only on mutation accumulation, but also on the ability of the immune system to recognize and eliminate abnormal cells before tumors become clinically significant. The adaptive immune system, particularly cytotoxic CD8+ T cells, plays an important role in immune surveillance by identifying tumor-associated antigens and destroying potentially dangerous cells. However, tumors can develop mechanisms that allow them to evade immune recognition, increasing the likelihood of continued growth and progression.

Recent research has also shown that immune function declines with age. Processes such as reduced T-cell turnover, decreased thymic output, and lower T-cell receptor diversity weaken adaptive immune surveillance over time. As immune efficiency decreases, the probability that tumors escape immune detection may increase as well. Because tumor growth and immune recognition both involve randomness and probabilistic biological interactions, stochastic modeling provides a useful framework for studying these dynamics.

This project was inspired by the stochastic tumor progression framework developed by George and Levine (2018), which modeled tumor growth and immune evasion using stochastic population dynamics. Rather than reproducing the full complexity of the original model, this project focused specifically on how immune strength, immune aging, and tumor growth rate influence tumor escape probability. A simplified stochastic simulation was developed in Python using Monte Carlo methods to investigate how varying immune detection probability and immune decline affect tumor progression outcomes.

The primary biological hypothesis of this project is that stronger immune surveillance reduces tumor escape probability, while increasing immune decline and faster tumor growth increase the likelihood of tumor escape over time.

Background: The adaptive immune system continuously monitors the body for abnormal cells through a process known as immune surveillance. Cytotoxic CD8+ T cells are capable of recognizing tumor-associated antigens presented on abnormal cells and eliminating those cells before tumors become clinically detectable. However, tumor cells are not passive targets and can develop immune-evasive phenotypes that reduce immune recognition and increase survival probability.

George and Levine modeled tumor progression as a stochastic birth process coupled with adaptive immune recognition. In their framework, tumor populations grow probabilistically while immune recognition occurs as a stochastic event dependent on immune functioning. Their

work demonstrated that declining immune efficiency substantially increases tumor escape probability and may help explain age-related cancer incidence patterns.

Stochastic modeling is particularly useful in biological systems because many cellular processes involve randomness. Tumor progression depends on probabilistic mutation events, variable cellular growth rates, and uncertain immune recognition. Unlike deterministic models, stochastic simulations can capture variability between individual tumor trajectories and account for rare but biologically meaningful escape events.

In this project, tumor growth was modeled using a stochastic Poisson birth process. Immune recognition was represented probabilistically using an immune detection parameter p , which determined the likelihood that the immune system successfully recognized and eliminated tumor cells during each simulation step. Aging was modeled through an exponential decline in immune detection probability over time.

Model Development and Approach: The simulation developed for this project was intentionally designed as a simplified version of the George and Levine framework. Rather than modeling multiple tumor populations or adaptive immune coevolution, the model focused specifically on the relationship between immune decline and tumor escape probability.

Each simulation began with a single tumor cell. At each time step, tumor growth occurred stochastically using a Poisson growth process. Tumor size therefore changed probabilistically rather than deterministically, reflecting the natural variability present in biological systems. During each time step, the immune system attempted to recognize and eliminate the tumor with probability p . If immune recognition occurred, the tumor was eliminated and the simulation ended. If the tumor population reached a predefined escape threshold before immune detection occurred, the tumor was considered to have escaped immune surveillance.

To model immune aging, immune detection probability gradually declined over time according to an exponential decay function. This simplification represented biological processes such as reduced adaptive immune efficiency, lower T-cell turnover, and declining immune surveillance associated with aging. The simulation algorithm followed these steps:

1. Start with a single tumor cell.
2. Simulate stochastic tumor growth using a Poisson process.
3. Apply probabilistic immune detection.
4. Eliminate the tumor if detection occurs.
5. Record tumor escape if the tumor reaches the escape threshold.
6. Repeat the simulation many times to estimate average escape probability.

Monte Carlo simulations were performed using repeated independent stochastic trials. Escape probability was estimated by averaging tumor outcomes across many simulations.

I did have to make some simplifying assumptions to focus the model specifically on immune decline and tumor escape probability. The model did not include spatial tumor structure, multiple immune cell populations, immune memory, or mutation-driven tumor evolution. Immune detection was represented using a single probability parameter rather than detailed immune signaling pathways. These simplifications allowed the project to focus directly on the relationship between immune strength, immune aging, and tumor escape dynamics.

Results: The first set of simulations examined how initial immune detection probability affected tumor escape probability. Results showed that stronger immune surveillance substantially reduced tumor escape probability. As immune detection probability increased, tumors were more likely to be eliminated before reaching the escape threshold.

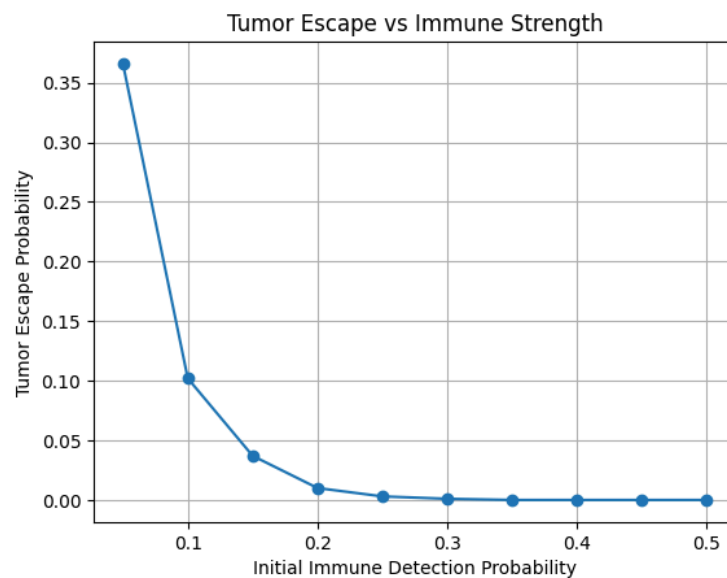


Figure 1. Tumor escape probability as a function of initial immune detection probability. Stronger immune surveillance reduced tumor escape probability by increasing the likelihood of early tumor elimination.

The second set of simulations investigated the effects of immune aging on tumor escape probability. Immune detection probability gradually declined over time according to an exponential decay parameter representing age-related immune decline. Results showed that increasing immune decline rates consistently increased tumor escape probability.

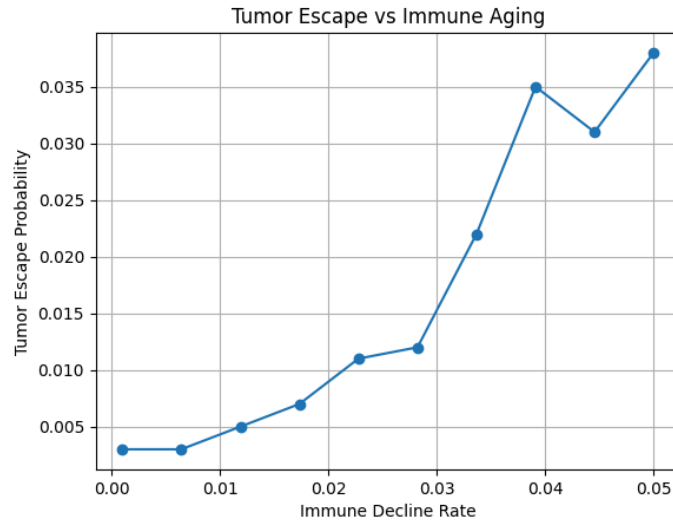


Figure 2. Tumor escape probability as a function of immune decline rate. Increasing immune decline resulted in higher tumor escape probability as immune function weakened over time.

An additional extension explored how tumor growth rate affected tumor escape dynamics. Faster tumor growth allowed tumors to reach the escape threshold more quickly, reducing the amount of time available for immune recognition. Simulations demonstrated that increasing tumor growth rate substantially increased escape probability.

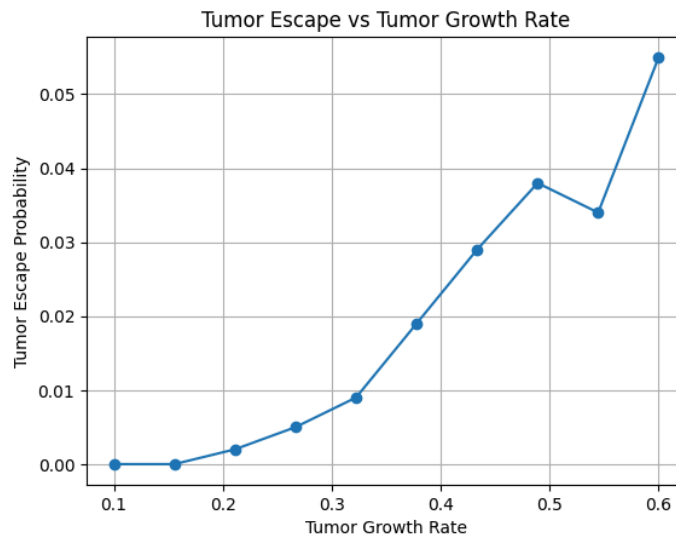


Figure 3. Tumor escape probability as a function of tumor growth rate. Faster-growing tumors showed higher escape probabilities because rapid proliferation reduced opportunities for immune detection.

Finally, a sensitivity analysis was performed to examine the combined effects of immune strength and immune aging. Multiple combinations of immune detection probability and immune decline rate were simulated to determine how these parameters interact to influence tumor progression outcomes. The results showed that low immune strength combined with rapid immune decline produced the highest probabilities of tumor escape.

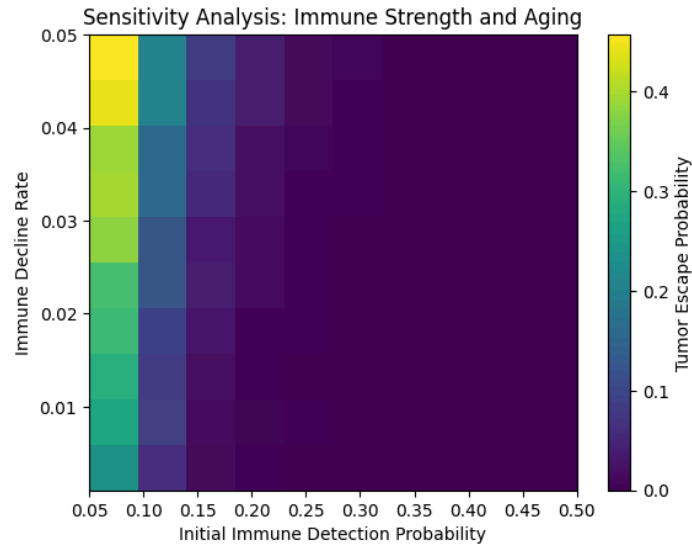


Figure 4. Sensitivity analysis heatmap showing tumor escape probability across varying levels of immune strength and immune decline. Lower immune strength combined with faster immune aging produced the highest tumor escape probabilities.

Overall, the simulations consistently demonstrated that stronger immune surveillance reduces tumor escape, while increasing immune decline and faster tumor growth increase the likelihood of tumor progression.

Discussion: One of the most important conclusions of the project is that declining immune function appears to substantially increase the likelihood of tumor escape. Even relatively small reductions in immune detection capability produced measurable increases in tumor escape probability within the simulations. These findings support the broader hypothesis that age-related immune decline may contribute significantly to increased cancer risk over time.

The project also demonstrates the usefulness of stochastic modeling in quantitative biology. Tumor growth and immune recognition are highly variable processes influenced by random biological interactions and probabilistic cellular behavior. Deterministic models often describe only average system behavior, while stochastic simulations can capture variability between individual tumor trajectories and rare escape events that may ultimately determine disease progression.

Despite these limitations, the model successfully reproduced biologically reasonable trends relating immune decline to tumor escape probability. The sensitivity analysis further demonstrated how combinations of weakened immune surveillance and aging can substantially alter tumor progression outcomes. Future work could improve the model by incorporating multiple tumor subpopulations, more detailed immune interactions, mutation-driven immune evasion, or experimentally derived biological parameters. Additional studies could also explore the effects of immunotherapy or compare deterministic and stochastic tumor growth dynamics.

Overall, this project demonstrates how simplified stochastic models can still provide meaningful insight into the relationship between immune surveillance, aging, and tumor progression.

George and Levine Paper Referenced: George, J. T., & Levine, H. (2018). Stochastic modeling of tumor progression and immune evasion. *Journal of Theoretical Biology*, 458, 148–155.

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